

DINESH A

Electrical & Electronics Engineering Student

CONTACT

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TECHNICAL SKILLS

- **Embedded & IoT:** Arduino, ESP32, Raspberry Pi, Sensors, GPS, GSM, Real-Time Systems
- **Hardware Design:** Verilog, VHDL, Digital Logic, ALU & Processor Architecture
- **Programming:** Java, C, C++, SQL, OOP, DSA
- **AI / ML:** Machine Learning, Artificial Intelligence, Data Preprocessing, AI Tools
- **Tools:** Git, GitHub, VS Code, MATLAB, Simulink, Arduino IDE, Raspberry Pi IDE, Antigravity IDE, OpenCode

LANGUAGES

- **English:** Professional Working Proficiency
- **Tamil:** Native

SOFT SKILLS

- **Team Collaboration**
- **Leadership**
- **Problem Solving**
- **Technical Communication**
- **Adaptability**
- **Time Management**

PROFILE

Electrical & Electronics Engineering student building toward **Embedded Systems, Product Development and R&D**, with hands-on experience in hardware prototyping, **IoT**, processor/GPU architecture simulation and intelligent systems. Combines Arduino/ESP32/Raspberry Pi embedded work with **Verilog/VHDL digital design**, a Java and Data Structures & Algorithms foundation, and applied **Machine Learning / AI**. Technology entrepreneurship through **NXRA INSIGHTS** adds product ideation, technical execution and cross-functional team experience.

EXPERIENCE

NXRA INSIGHTS Pvt Ltd — Director, Technology & Product Development *(Mar 2025 – May 2026)*
- Led technology/product ideation, technical planning and hardware/software solution design across embedded systems, IoT, automation and prototyping initiatives with an 8-member technical team.
- Coordinated cross-functional troubleshooting, technical documentation and iterative prototype development under resource constraints.

Industrial Intern — Daikin Air Conditioning India Pvt Ltd, Sricity, Andhra Pradesh *(30 Jun – 4 Jul 2025)*
- Completed an HVAC Skill Enhancement industrial training program covering air-conditioning fundamentals, R-32 refrigerant, inverter ACs, VRV systems and installation basics.

Industrial Intern — Alfa Switchgear (I) Pvt Ltd, Erode *(23–27 Dec 2024)*
- Completed internship training in LV/MV switchgear panel manufacturing (up to 36kV), gaining exposure to industrial electrical engineering practices.

AI/ML Intern — QMOS Technologies Pvt Ltd *(15–31 Dec 2025)*
- Worked on machine learning workflows including data preprocessing, model development and evaluation.

SELECTED PROJECTS

Simple GPU Architecture Simulation (Team) — GPU scheduling, SM status visualization, scheduler logic and live waveform monitoring. Team: Al Zafirah M, Baskar Raja S, Dharanishwaran S; Guide: Mr. M. K. Anandkumar, M.E., (Ph.D.).
AI Chip Simulator — Digital Design / Processor Architecture — Modeled ALU, control unit, registers and instruction execution; verified logic in Verilog/VHDL, simulating fetch-decode-execute behavior.
Smart Safety Helmet — IoT & Embedded Systems — Helmet-use, alcohol and accident detection with GPS/GSM emergency communication and safety monitoring.
Raspberry Pi Fingerprint Attendance/Access System — Raspberry Pi 3 Model B+, ESSL X990-C fingerprint scanner, 2.4" TFT/SPI display; device interfacing, Linux/Python setup, power-backup considerations.
Additional Projects — Vision Buddy, Smart Ration Queue, Pet Tracking System, Animal Detection (ESP32), Auto Security Detection System, Workplace Security System, Email Spam Detection, Agriculture Monitoring System, Smart Pill Box.

EDUCATION

B.E. Electrical and Electronics Engineering — Excel Engineering College (Autonomous) | Expected 2027
Relevant Coursework: Data Structures & Algorithms, Digital Electronics, Microprocessors & Microcontrollers, Machine Learning, Embedded Systems, OOP

CERTIFICATIONS

- **MathWorks Training Services — 100% Completion:** Battery State Estimation, MATLAB Onramp, Simulink Onramp, Simscape Onramp.
- **NPTEL, IIT Kharagpur (Elite):** Programming in Java (72%), Introduction to Internet of Things (69%).
- **Great Learning Academy:** Introduction to Artificial Intelligence — Completed, December 2023.
- **Infosys Springboard / LinkedIn Learning — JavaScript:** Getting Started with JavaScript Programming; Functions & Arrays.

LEADERSHIP & ACTIVITIES

- **IEEE Member (Madras Section) | Student Branch Webmaster:** Supported the branch's technical web presence and student engagement activities.
- **Olimayam 3.0 — Pitch Smart Bootcamp & Finale, StartupTN:** Solo participant; received StartupTN "Start Step" entrepreneur recognition.
- **IEEE Xtreme 18.0 Programming Competition:** Participated with Team RATHAN in the 24-hour global contest.
- **IDE Bootcamp — PSNA College, AICTE/MoE Innovation Cell:** Participated, Edition 2 Phase II, Innovation Design & Entrepreneurship.
- **GUVI Interior Tech Gala Workshop:** Participated, Excel Engineering College.
- **ELYTRON'26 Hackathon:** Participated, Excel Engineering College.
- **Ongoing Participation:** Hackathons, technical workshops and paper presentations, frequently in a coordinating/leading capacity.